

Reaching the Actionable Radius: Physics-Guided Posterior Distillation for Methane Source Attribution

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Abstract—We present a simple, drop-in training recipe that brings calibrated machine-learning attribution of methane super-emitters under the U.S. EPA’s 50m actionability radius — the precision at which a detection becomes a repair. On a methane-configured testbed whose closed-form physics makes the *exact* Bayesian posterior computable for every scenario, we first contribute an audit that measures, in meters, how much of the physically available information a calibrated localizer delivers: a conformal-wrapped deep localizer with a verified 90% coverage guarantee achieves a 94m median search radius where the data support 7m. Guided by a new one-run diagnostic — conformalizing the distillation *teacher* itself — we localize the opportunity: the missing ingredient is a classical computation (per-cell matched filtering) that pooled set encoders cannot represent, rather than better supervision or a more modern density estimator. Our recipe supplies it: two closed-form matched-filter maps as input channels, a zero-initialized residual head (0.2% extra parameters), and distillation from the tempered exact posterior. Across three architecture families and nine seed-runs, every model reaches 43.6–49.3m at unchanged coverage and millisecond latency, outperforming a fairness-hardened flow-based neural posterior estimator (55.5–71.2m). Under realistic wind-measurement error, our distilled students retain calibration where exact-physics inversion cannot (0.89–0.91 vs. 0.70 coverage), and uncertainty-aware input maps recover a further 12m — evidence that learned, physics-guided attribution is not only sharper but more robust than the physics alone.

Index Terms—methane monitoring, conformal prediction, simulation-based inference, knowledge distillation, sensor networks, physics-guided machine learning

I. INTRODUCTION

Methane delivers roughly $80\times$ the 20-year warming potency of CO_2 and decays within about a decade, making rapid repair of oil-and-gas *super-emitters* one of the highest-leverage near-term climate actions. Detection has advanced rapidly — UNEP’s satellite alert system issued over a thousand notifications in its first two years [1] — and the remaining step is *attribution*: converting a detection into a source location trustworthy enough to dispatch a crew. Regulation makes the target concrete: the EPA Super-Emitter Program operationalizes actionability as a 50m radius [2], with analogous source-level duties in the EU [3]. Continuous monitoring with sparse networks of fixed point sensors is the tier of the monitoring stack where this can happen: a handful of cheap masts around a facility, always on, catching the plume as it sweeps across them under naturally meandering wind.

Machine learning supplies the speed, and conformal prediction [4] supplies a guarantee: a neural localizer maps readings to a probability map, and a split-conformal wrapper converts it into a region that provably contains the source 90% of the time. **The gap** is that coverage guarantees frequency, not usefulness. Two questions had no quantitative answer: (i) how close is a calibrated region to the *smallest* region the sensor data physically support, and (ii) if there is a gap, what mechanism causes it and what closes it? Existing diagnostics for amortized inference check whether a posterior is well-calibrated [5], not how far it is from the exact posterior at matched calibration — a question that requires exact posteriors to answer, and that determines whether ML monitoring meets the regulatory bar.

Contributions. We answer both questions constructively:

- **C1 — An exact-posterior audit.** A methane testbed (CH₄-T) whose closed-form physics makes the true Bayesian posterior of every scenario computable, so the efficiency of any calibrated localizer can be measured in meters against the information limit (7m) and the regulatory radius (50m).
- **C2 — A diagnosis, by elimination, with a reusable tool.** Conformalizing the distillation *teacher* decomposes the gap into a supervision ceiling (11.7m) versus student approximation (94m), and controlled tests of four rival explanations identify the true bottleneck: a classical computation the encoder cannot represent.
- **C3 — A recipe that reaches the bar.** Physics-guided posterior distillation — matched-filter input maps, a zero-initialized residual head, and a tempered exact-posterior teacher — brings three different architecture families to 43.6–49.3m at unchanged coverage and latency, and remains calibrated under wind-measurement error that defeats exact-physics inversion.

The components are deliberately classical — matched filters are 1940s detection theory; distillation and conformal prediction are standard — and we verify empirically that this is a strength: the composition outperforms the current flow-based state of the art at identical information.

II. RELATED WORK

Conformal prediction provides distribution-free coverage [4]; efficiency (region size) is recognized as the open half of the problem. We contribute the first audit we are aware of that

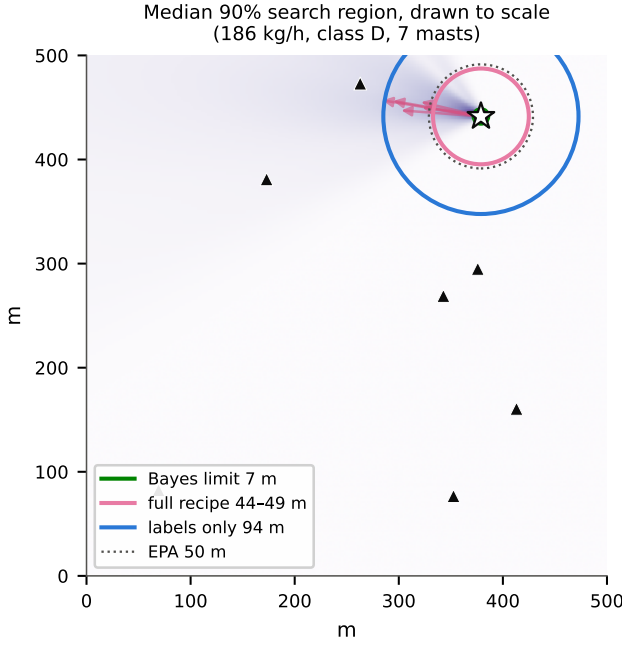


Fig. 1. The opportunity, to scale. A wind-swept methane plume (purple; meander arrows), seven masts (black), and the true source (star) on a 500 m site. The sensor data support a 7 m search circle (green); a standard calibrated localizer delivers 94 m (blue); our recipe reaches 44–49 m (magenta), inside the EPA 50 m actionability radius (dotted).

scores conformal efficiency against the *exact* Bayesian limit at matched, verified coverage. **Simulation-based inference** amortizes posteriors with neural density estimators [6], [7], benchmarked at simulation budgets matching ours [8]; its standard diagnostics test calibration [5] but cannot measure distance from the exact posterior — our audit asks precisely that, and our flow-based baseline places the current SBI standard *inside* the comparison. **Source-term estimation** inverts dispersion models via optimization or MCMC at minutes-to-hours per event [10]; deep localizers achieve millisecond inference but have lacked instruments to certify how much information they extract. **Physics-guided ML and distillation** inject domain structure through features, losses, or teachers [9]; our teacher is an exact Bayesian oracle rather than a larger network (compressing inference cost, not parameters), and our input maps are the forward model’s sufficient statistics. **Controlled-release evaluation** (METEC) scores continuous monitoring systems on detection, quantification, and equipment-scale localization [11]; our testbed mirrors that task structure at super-emitter rates with exact ground truth, complementing — not replacing — field validation.

III. METHODOLOGY

A. Testbed and exact-posterior audit (C1)

CH4-T places a source of unknown location and rate ($10\text{--}500\text{ kg h}^{-1}$, log-uniform, bracketing the 100 kg h^{-1} super-emitter threshold) on a $500\times 500\text{ m}$ site sensed by 4–12 masts (ppm readings, 0.2–3 ppm noise, dropout). The plume

is a steady Gaussian under known Pasquill–Gifford stability with Briggs open-country dispersion (coefficients verified against published tables; mass flux conserved to 4×10^{-6}), driven quasi-statically by a known fluctuating wind record ($1\text{--}8\text{ m s}^{-1}$, AR(1) meander; steps read as 1-min averages, a 30-min window). The veering wind sweeps the plume across the masts; this time-of-arrival information makes 94% of scenarios well-identifiable (vs. 43% under constant wind).

Because the response is closed-form and linear in rate, the exact posterior over a 64×64 grid is computable per scenario (adaptive rate quadrature, float64, normalization error $\sim 10^{-12}$). A blocking gate requires conformal prediction applied to the exact posterior itself to self-calibrate (0.95 observed) — a gate that caught two implementation bugs during development. All models share identical data (10k/1k/2k/2k train/val/calib/test, the modal SBI budget [8]), identical D4 symmetry augmentation, and an identical randomized-HPD split-conformal wrapper at verified 90% coverage in numerically safe tail-mass form. Region sizes are reported as equal-area circle radii.

B. Locating the opportunity (C2)

A DeepSets localizer trained on one-hot labels is calibrated (coverage 0.89) at a 94 m median radius — the information for 7 m is in the readings, waiting to be extracted. Distilling the exact posterior helps (74 m) and reveals structure: the raw near-delta posterior must be tempered by a 0.75-cell blur to be learnable (raw: 134 m). Our diagnostic then asks how much the tempered teacher itself supports: running the same conformal audit *on the teacher* yields 11.7 m — so supervision content permits a $6\times$ sharper student, and the remaining gap is approximation. Controlled tests eliminate the remaining suspects: sharper targets (anneal: 87 m), a geometry-aware Sinkhorn transport loss (243 m; implementation-scoped), and the current SBI-standard flow head — given a $2.5\times$ budget, a widened head, and full-band conditioning after adversarial review of our own implementation — improves the labels-only baseline to 62.8–71.2 m, and we verify its spline head can express 4.4 m regions, so its remaining blur is not head-structural. What survives elimination: localization requires correlating each of 4,096 candidate cells’ space–time signature against all readings — *per-cell matched filtering* — which a pooled set representation cannot carry. The opportunity is to supply that computation.

C. The recipe (C3)

For any base localizer (Fig. 2): **(1)** compute per cell c the sufficient statistics $a_c = \sum g_c y$, $b_c = \sum g_c^2$ of the linear-Gaussian model (g_c the unit-rate response) and feed two 64×64 channels, the matched-filter statistic $z_c = a_c / (\sigma \sqrt{b_c})$ and $\log b_c$ — milliseconds, closed form, computed from the network’s own inputs (no new information, only a computation); **(2)** add a 3-layer zero-initialized conv head ($\sim 10\text{k}$ parameters, 0.2% of the base) summed with the localizer’s logits, making every comparison controlled by construction; **(3)** train with cross-entropy to the tempered exact posterior —

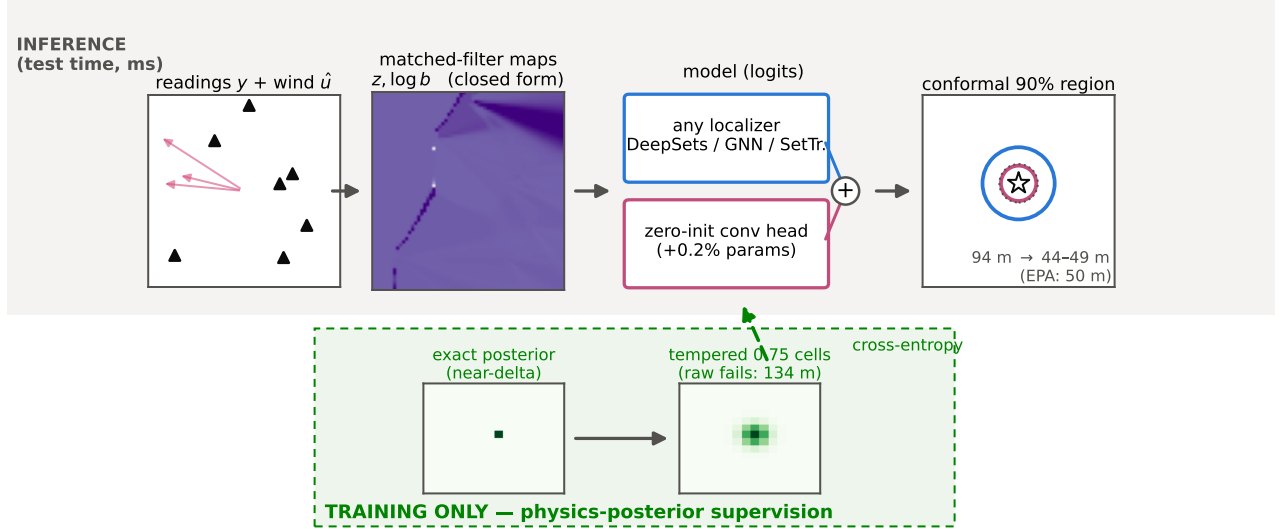


Fig. 2. Physics-guided posterior distillation, on real artifacts from one test scenario. Top band — the inference path, milliseconds at test time: (1) observed mast readings and wind record; (2) two closed-form physics maps computed from the same observations (the classical matched-filter statistic per candidate cell and a sensitivity map — sufficient statistics of the forward model); (3) any localizer plus a zero-initialized convolutional head reading the maps: at step 0 the model is the unmodified baseline, and training opens the physics channel only where it reduces loss; then a split-conformal wrapper certifies 90% coverage — the median region reaches 44–49 m against the EPA 50 m radius. Bottom band — training only: targets are the exact Bayesian posterior of each scenario, tempered by a 0.75-cell blur; no inference cost.

TABLE I

MEDIAN 90% CONFORMAL REGION RADIUS (M); RANGE OVER 3 SEEDS; COVERAGE 0.89–0.91 IN EVERY CELL. REFERENCES: BAYES LIMIT 7.0 M, TEACHER CEILING 11.7 M, EPA ACTIONABILITY RADIUS 50 M.

Encoder	Labels	+Distill	+Maps (full)
DeepSets	94–98	73–74	48.1–49.3
GNN	55–58	49–52	43.6–44.3
Set Transformer	69–75	56–60	44.7–48.9

TABLE II

THE CURRENT SBI-STANDARD HEAD ON THE SAME ENCODER, DATA, AND AUDIT; 3 SEEDS PER ROW.

Model (DeepSets encoder)	Radius (m)	Coverage
Flow-NPE (fairness-hardened)	62.8–71.2	0.90–0.91
Flow-NPE + maps + distillation	55.5–56.4	0.89–0.90
Ours (grid head, full recipe)	48.1–49.3	0.89

the maps supply evidence, the teacher teaches how to weigh evidence into calibrated probability; (4) wrap with unchanged split conformal.

Why amortize when the teacher exists? On this testbed the teacher is the measuring instrument, tractable by construction. In deployment, realistic dispersion has no closed form and simulator inversion costs hours per event [10] where monitoring needs seconds — and, as §IV-C shows, the distilled student holds calibration under sensor-measurement error that exact inversion does not survive. Every simulated training set already contains its own uncertainty labels; this recipe is how to extract them.

IV. RESULTS

A. One recipe, three architectures

Every ingredient improves every architecture, and all nine full-recipe seed-runs land under the 50 m bar (Table I; Fig. 3A). Paired Wilcoxon on per-scenario log region sizes: $p < 10^{-200}$. Two further findings strengthen the result. First, the spread across architectures collapses (94-vs-56 without the

recipe; 49-vs-44 with it): supervision and inputs, not encoder design, were the binding constraint, so gains are available to whatever architecture a practitioner already has. Second, point accuracy is essentially unchanged — the recipe improves the *shape of uncertainty*, which is exactly what region size measures and what dispatch decisions need.

B. Ahead of the modern standard

The flow head is a genuine advance — it outperforms plain distillation with no physics at all — and our levers transfer onto it monotonically (Table II). The full recipe stays ahead: a convolutional head reading the maps *spatially* extracts more than the flow’s parametric conditioning at identical information. Our approach is thus compared against, and complements, the current standard rather than being superseded by it.

C. Robust where exact physics is not

Real anemometers err. Under a $10^\circ/10\%$ observation error, exact inversion assigns numerically zero probability to the true source in 30% of scenarios, and no calibration procedure can

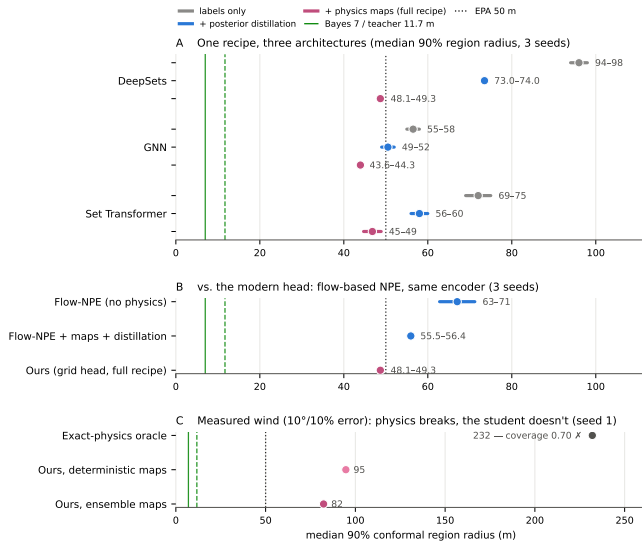


Fig. 3. All results on one axis. (A) The recipe across encoders; (B) the flow-head comparison; (C) measured-wind robustness. Vertical lines: Bayes limit 7 m (green), teacher ceiling 11.7 m (dashed), EPA 50 m (dotted). Whiskers span 3 seeds.

TABLE III

WIND MEASURED WITH REALISTIC ERROR (10° DIRECTION, 10% SPEED):
LEARNED STUDENTS RETAIN CALIBRATION; MODELING INPUT
UNCERTAINTY RECOVERS SHARPNESS.

Pipeline	Coverage	Radius
Exact-physics oracle	0.70	232 m
Ours, deterministic maps	0.91	94.7 m
Ours, ensemble maps	0.89	82.3 m

repair a zero — its conformal coverage falls to 0.70 (Table III). The distilled student, trained through the same noise, *retains* its guarantee: a softmax never declares a location impossible, and training widens uncertainty exactly where wind error demands. Making the physics inputs uncertainty-aware — *ensemble maps*, the mean and spread of the matched-filter maps over eight draws from the wind-error model — recovers 12 m ($p=10^{-122}$). Amortization thus delivers robustness the exact physics cannot: the student outlives its own teacher, and modeling input uncertainty converts measurement error from a failure mode into a quantified, budgeted cost.

V. LIMITATIONS

CH4-T is an instrument, not a deployment: single steady sources (no zero-leak or multi-source scenarios), flat terrain without equipment wakes, known stability class, quasi-static stepping, in-distribution marginal coverage. Rates sit in the super-emitter regime, above the $\leq 7 \text{ kg h}^{-1}$ range of standard METEC continuous-monitoring tests [11]; localization granularity matches what such facilities score. The residual factor of ~ 6 to the 7 m limit is open headroom; the transport-loss null is implementation-scoped; a wind-marginalized Bayes limit under measurement error is future work (the marginalization factorizes over timesteps; reliable quadrature requires

bearing-adaptive panels). Field claims await controlled-release validation — and every comparison reported here is exact regardless, because all models share one verified simulator: fidelity bounds transfer, not the tables.

VI. CONCLUSIONS

Three numbers organize this work: the physics allows 7 m, regulators ask for 50 m, and a standard calibrated localizer delivers 94 m. We contribute the instrument that measures this gap (C1), the diagnosis that locates its mechanism — a classical computation, not a learning deficiency (C2) — and a deliberately simple recipe that closes it on every architecture tested, at unchanged cost, while remaining calibrated under the sensor error that defeats exact inversion (C3). The broader message is constructive: simulated training data already contain their own uncertainty labels, classical signal processing already knows computations networks need not rediscover, and combining the two turns a coverage guarantee into an actionable one. We hope the evidence standard — exact-posterior audits, blocking self-calibration gates, one-change-at-a-time controls with documented nulls — travels furthest of all.

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